Nano Dosimetry Meeting

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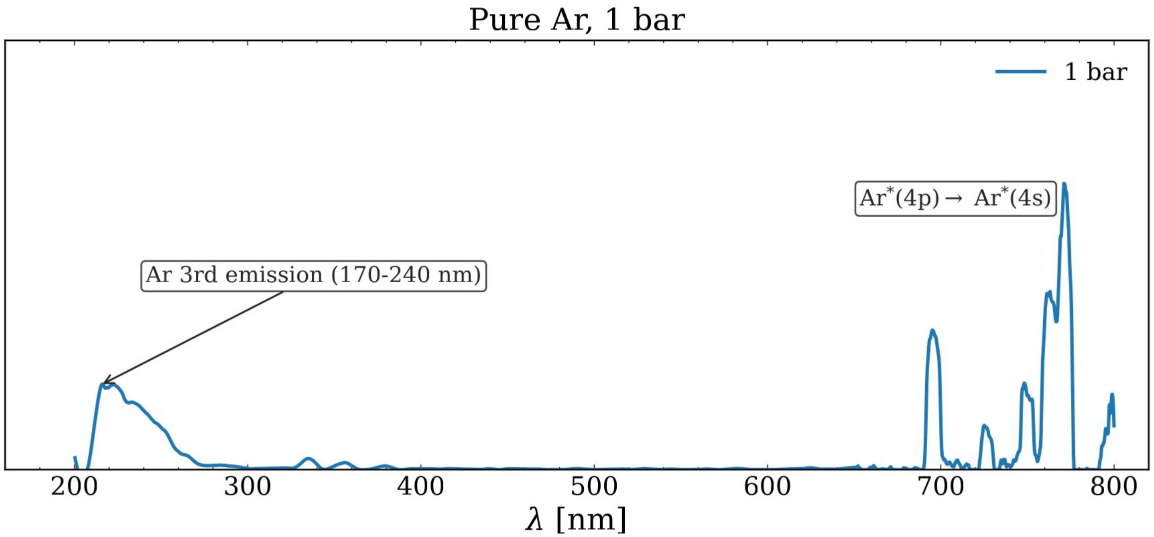
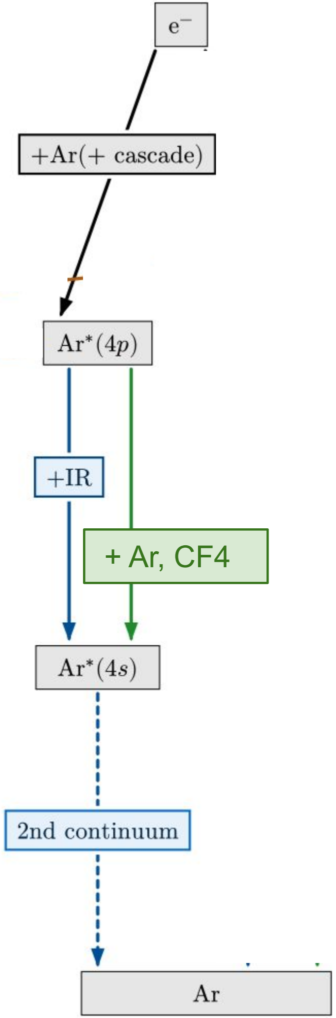
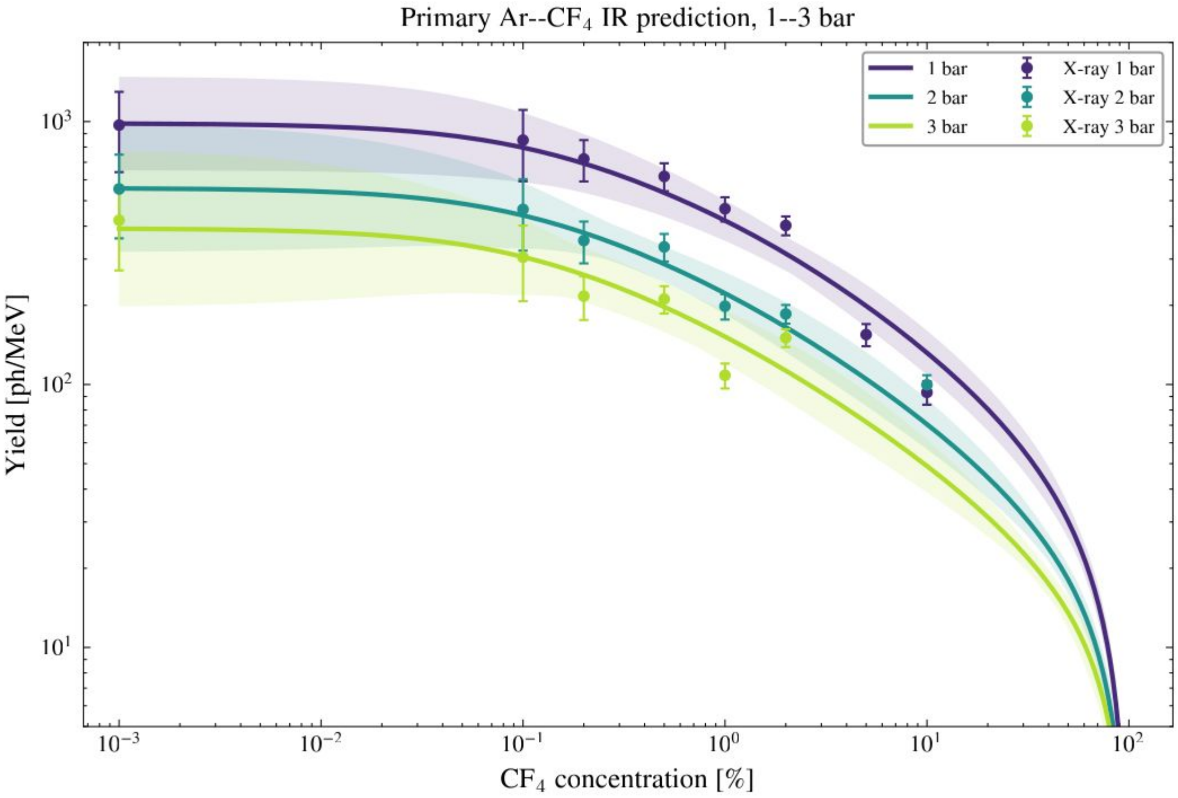
Model

$$Y_{\mathrm{IR},\lambda} = N_{\mathrm{Ar}^{**},\lambda} \cdot \mathcal{W}_{\mathrm{Ar}^{**},\lambda} \cdot \left(\frac{1/\tau_{\mathrm{Ar}^{**},\lambda}}{1/\tau_{\mathrm{Ar}^{**},\lambda} + n \cdot f \cdot K_{\mathrm{Ar}^{**},Q(\mathrm{Ar}),\lambda} + n \cdot (1-f) \cdot K_{\mathrm{Ar}^{**},Q(\mathrm{CF}_4),\lambda}} \right)$$



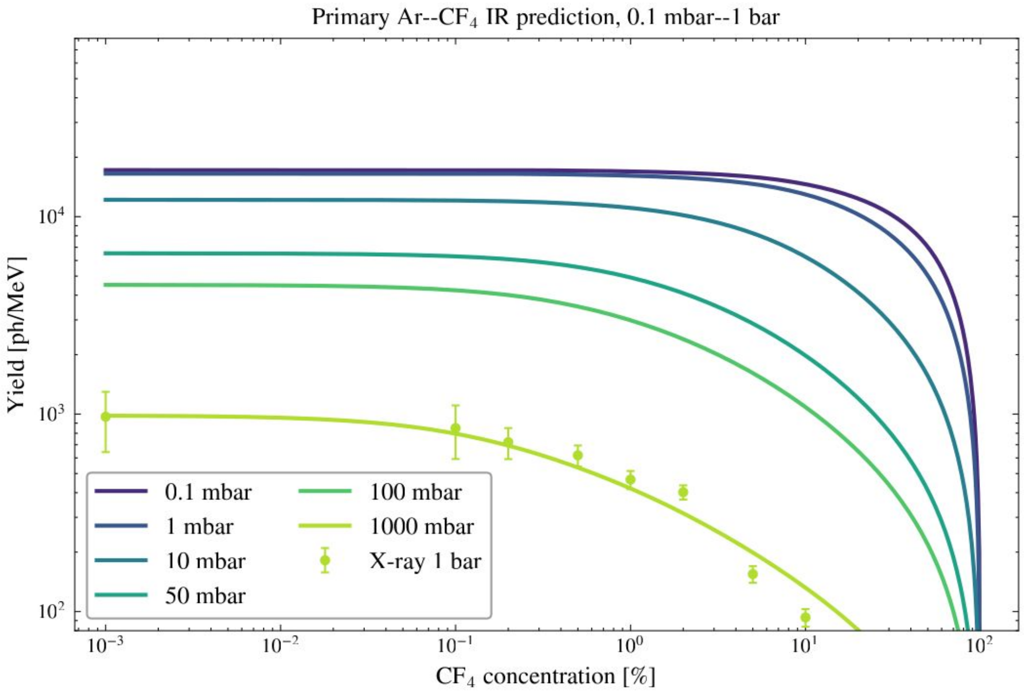


Predictions 1-3 bar



Predictions 1bar - 1mbar

Predicción	Valor
$Y_{\mathrm{IR}}^{\mathrm{ArCF_4}}(100\,\%\ \mathrm{Ar},\ 0.1\,\mathrm{mbar})$	17 000
$Y_{\mathrm{IR}}^{\mathrm{ArCF_4}}(100\,\%\ \mathrm{Ar},\ 1\,\mathrm{mbar})$	17 000
$Y_{\mathrm{IR}}^{\mathrm{ArCF_4}}(100\,\%\ \mathrm{Ar},\ 10\,\mathrm{mbar})$	12 000
$Y_{\mathrm{IR}}^{\mathrm{ArCF_4}}(100\,\%\ \mathrm{Ar},\ 50\,\mathrm{mbar})$	6500
$Y_{\mathrm{IR}}^{\mathrm{ArCF_4}}(100\,\%\ \mathrm{Ar},\ 100\,\mathrm{mbar})$	4500
$Y_{\mathrm{IR}}^{\mathrm{ArCF_4}}(100\,\%\ \mathrm{Ar},\ 1000\,\mathrm{mbar})$	980



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Comparison with literature

Determinación	$K_{Ar^{**}Q(Ar)}$	$K_{Ar^{**}Q(CF_4)}$
Sadeghi et al. [42]	1.63×10^{-17}	1.80×10^{-16}
Ar–CF_4 UV/VIS	1.17×10^{-17}	4.28×10^{-16}
Ar–CF_4 IR, promedio	3.69×10^{-17}	1.30×10^{-15}

VUV

Predicción	Valor
$Y_{\mathrm{Ar2nd,total}}(100\ \%\ \mathrm{Ar})$	17 000
$Y_{\mathrm{CF_4^+(D\rightarrow X)}}(100\ \%\ \mathrm{CF_4})$	430